Sequences and induction

Lecture and worked examples

Objectives

Compute terms and a sum; write an induction proof; establish monotonicity and bounds; use an auxiliary sequence.

Defining a sequence and computing

A real sequence is a function defined on $\mathbb{N}$, or on the integers from some index n_0 onwards. We write u_n for its term of index n. An explicit formula gives u_n directly; a recurrence relation also requires an initial term. For example, $u_0 = 2$ and $u_{n+1} = 3u_n - 1$ give $u_1 = 5$, $u_2 = 14$.

Defining a sequence and computing

If $u_{n+1} = u_n + r$, then $u_n = u_p + (n-p)r$. If $u_{n+1} = qu_n$, then $u_n = u_p q^{n-p}$ for $n \geq p$.

$$\sum_{k=p}^{n} u_k = (n-p+1)\frac{u_p + u_n}{2} \quad \text{(arithmétique)}, \qquad \sum_{k=p}^{n} u_k = u_p \frac{1-q^{n-p+1}}{1-q} \quad (q \neq 1).$$

When $q = 1$, the second sum equals $(n-p+1)u_p$.

Proof by induction

To prove $P(n)$ for every integer $n \geq n_0$: check $P(n_0)$; fix an arbitrary integer $n \geq n_0$ and assume $P(n)$ is true; deduce $P(n+1)$; conclude by the principle of induction. The hypothesis is not the conclusion: it may only be used at the fixed index.

Proof by induction

Let $u_0 = 1$, $u_{n+1} = 2u_n + 3$. For $n \in \mathbb{N}$, let $P(n)$ be: "$u_n = 4 \times 2^n - 3$".

Base case. $4 \times 2^0 - 3 = 1 = u_0$; $P(0)$ is true.

Inductive step. Fix $n \in \mathbb{N}$ and assume $P(n)$ is true. By the definition of the sequence, then by this hypothesis,

$$\begin{aligned} u_{n+1} &= 2u_n + 3 \\ &= 2(4 \times 2^n - 3) + 3 \\ &= 8 \times 2^n - 3 = 4 \times 2^{n+1} - 3. \end{aligned}$$

This is $P(n+1)$: the property is inherited.

Conclusion. By the principle of induction, $u_n = 4 \times 2^n - 3$ for every $n \in \mathbb{N}$.

Proof by induction

Let $P(n)$ be: "$\sum_{k=0}^{n} k = \frac{n(n+1)}{2}$". For $n = 0$, both sides equal 0: $P(0)$ is true. Fix $n \geq 0$ and assume $P(n)$ is true. Then

$$\begin{aligned}\sum_{k=0}^{n+1} k &= \sum_{k=0}^{n} k + (n+1) \\ &= \frac{n(n+1)}{2} + (n+1) \\ &= \frac{(n+1)(n+2)}{2}.\end{aligned}$$

We have established $P(n+1)$. By induction, the formula holds for every $n \in \mathbb{N}$.

Global behaviour

A sequence is increasing if $u_{n+1} \geq u_n$ for every index, decreasing if the inequality is reversed. It is bounded above if there is a real number M such that $u_n \leq M$ for every n; bounded below if there is m such that $u_n \geq m$; bounded if it is both bounded below and bounded above. To study monotonicity, one can study the sign of $u_{n+1} - u_n$. If all terms are strictly positive, one can compare $\frac{u_{n+1}}{u_n}$ with 1. Studying a function f such that $u_n = f(n)$ is another possibility.

Global behaviour

For $u_{n+1} = au_n + b$, $a \neq 1$, solve $\ell = a\ell + b$, then set $v_n = u_n - \ell$. This gives $v_{n+1} = av_n$, hence

$$u_n = \ell + (u_0 - \ell)a^n, \qquad \ell = \frac{b}{1-a}.$$

If $a = 1$, the sequence is arithmetic: do not divide by $1 - a$.

Global behaviour

If $u_0 = 10$ and $u_{n+1} = 0.8u_n + 6$, then $\ell = 30$ and $u_n = 30 - 20(0.8)^n$. We have $10 \leq u_n < 30$ and $u_{n+1} - u_n = 4(0.8)^n > 0$.

Global behaviour

Checking a property on a hundred terms does not prove it. An increasing sequence is not necessarily positive; a bounded sequence is not necessarily monotonic.

Research activity

A tank topped up every day - 35 min.

A tank initially contains 100 litres of water. Each day, 20 % of the volume present is drawn off, then 30 litres are added. For every natural number n, let u_n be the volume of water, in litres, after these two operations have been carried out n times. Thus $u_0 = 100$.

Research activity (2)

1. Compute u_1 and u_2, then express u_{n+1} in terms of u_n.
2. Which volume would remain unchanged after one day? Prove by induction that, for every $n \in \mathbb{N}$, $100 \leq u_n < 150$.
3. For every $n \in \mathbb{N}$, let $v_n = u_n - 150$. Show that (v_n) is geometric. Deduce the expressions of v_n and u_n in terms of n.
4. Write a Python function with no argument that returns the first integer n for which $u_n > 145$. Run the program and interpret the result.

Research activity (3)

5. A student has computed the first hundred terms and notes that they are all less than 150. Is this observation enough to prove that all the terms are? Justify.